MECH 191 — Metallurgy

Course Introduction

Welcome & Syllabus Overview

Fall 2026 · Mondays, 5:00 – 7:30 PM · Kai Santos

Welcome to MECH 191

Course Description

A comprehensive introduction to the principles of metallurgy and material science, focusing on the structure, properties, and performance of metals and alloys. Topics include mechanical testing, metallography, phase diagrams, heat treatment, mechanical properties, corrosion, and failure analysis.

You'll gain hands-on experience in material testing, alloy composition, and microstructural analysis, and learn how to select and apply materials for real industrial purposes.

Learning Objectives

- Explain the relationship between structure, properties, and performance of metals and alloys
- Describe the phases in a metal alloy system using a phase diagram
- Apply heat treatment processes to modify metal properties for engineering applications
- Conduct lab tests to measure hardness, tensile strength, and ductility
- Interpret failure mechanisms — corrosion and fatigue — from properties and usage conditions
- Assess metal/alloy suitability for engineering scenarios: performance, durability, cost

Course Logistics

Credits & Prerequisites

3 credits · No prerequisites

Dates

August 24 – December 18, 2026

(17-week semester)

Meeting Time

Mondays, 5:00 PM – 7:30 PM

Classroom

PS101B

Office Hours

By appointment only

Instructor

Kainani “Kai” Santos

kainanis@hawaii.edu

Content drawn from Moniz, Metallurgy (5th ed.) and Kalpakjian, Manufacturing Processes for Engineering Materials (6th ed.) — all materials provided digitally.

Grading Scheme

Final Exam

PS101B · December 18, 2026

5:00 – 7:00 PM

Must be returned by 7:00 PM or the score is automatically zero.

Final letter grades follow the standard A–F scale.

The Metallography Portfolio (50 pts) is graded internally across all 15 weekly modules plus a holistic end-of-semester review — see the Metallography Lab slide for details.

Course Roadmap — Five Units

Unit 1	Weeks 1–2	Foundations: What Metals Are and How They're Built
Unit 2	Weeks 4–6	Metal Families: Knowing What You're Working With
Unit 3	Weeks 7–9	Testing, Inspection & Quality: The Technician's Core Skills
Unit 4	Weeks 10–13	Manufacturing Processes: How Parts Are Made and What to Check
Unit 5	Weeks 14–16	Keeping Metal in Service: Welding, Failure, Corrosion & Modern Tech

Course Timeline

Tentative schedule — subject to change. Updates will be announced by the instructor.

The Metallography Lab

Every instructional week, you'll complete one metallography module using the shared class specimen kit and the EXAMET-5-R-600-HD microscope — examine a specimen, complete an observation table, sketch or photograph the key feature, and answer short-answer questions.

Check Out

See the instructor after class and sign out the specimen tray.

Examine

Work through the module's magnifications on the EXAMET-5, following the setup steps.

Submit

Return the kit before leaving class. All 15 modules are compiled into one portfolio, due at the end of the semester.

Course Policies

Generative AI Policy

No form of generative AI (ChatGPT, Claude, Microsoft Copilot, Apple Intelligence, etc.) may be used, in whole or in part, to generate course materials or assignments. Grammar/spell-check tools (e.g., Grammarly, Word, Google Docs) are permitted to refine your own writing.

Academic Integrity

Cheating and plagiarism violate the Student Conduct Code and can result in expulsion. See the Leeward CC academic dishonesty policy for full definitions.

Participation Verification

Students who don't participate in Week 1 are marked “No Show” and administratively disenrolled with a full refund.

Title IX

Leeward CC prohibits sex- and gender-based discrimination. Contact the Title IX Coordinator at leetix@hawaii.edu for resources and support.

Technology & Materials

Textbooks (Provided Digitally)

Metallurgy, 5th ed.

B.J. Moniz

Manufacturing Processes for Engineering Materials, 6th ed.

S. Kalpakjian

All content is provided as PDFs / video links. Students needing a different format (e.g., printed hardcopy) should make their own accommodations.

What You'll Need

- University-provided Gmail account
- Access to the learning management system (Lamaku)
- Reliable internet access outside class meeting times
- A laptop for in-class and take-home work

Coming Up — Week 1

Atomic Structure & Bonding

The atom, the periodic table, atomic bonding, alloys and solid solutions, crystal structures, and grain boundaries.

Questions? Reach out anytime.

Kainani “Kai” Santos · kainanis@hawaii.edu · CE401 · Office hours by appointment